

Scalability & Future Plan

Date	1 October 2026
Team ID	6ab224f9f32d8c9d8b2253f9
Project Name	Import data using transform maps
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Imports are started manually and cannot run on a schedule.	An admin must run every repeat load.	Medium
2	No email notification when an import has errors.	Errors may be noticed late.	Medium
3	Files of more than 5,000 rows have not been tested.	Large imports may be slow or time out.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	One admin runs the imports manually.	Give import roles to more users and run scheduled imports so no one has to start them.
2	Data Storage	Import Set staging tables keep the rows of every run.	Delete old Import Sets regularly and keep only the history that is needed.
3	Performance	Tested with files of up to 5,000 rows.	Split large files into batches and run big imports at off-peak hours.
4	Security	Access is limited with roles and ACLs.	Review roles regularly and keep the import tables limited to import admins.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Data Source, Import Set, Transform Map with coalesce, and validation script	Completed	Data is imported accurately without duplicates.
Phase 2	Scheduled imports and email notification on errors	Short term	Repeat loads run automatically and errors are noticed quickly.
Phase 3	Support for other data sources such as JDBC and REST	Medium term	Data can be imported straight from other systems without files.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 4	Reusable transform map templates and an import results dashboard	Long term	Faster setup of new imports and a clear view of import results.